Historical Data on the Photoelectric Effect: Lenard (1902) through Millikan (1916)

Background notes for T. J. Sargent

March 23, 2026

Contents

1 Overview

The photoelectric effect—the emission of electrons from a metal surface irradiated by light—was discovered by Heinrich Hertz in 1887. The subsequent two decades of experimental work culminated in Einstein's 1905 quantum hypothesis and its direct verification by Millikan in 1916. The central observable is the *stopping potential* V_0: the minimum retarding voltage needed to prevent the most energetic emitted electrons from reaching the collector. Einstein's equation predicts

$$V_0 = \frac{h}{e}\,\nu - \frac{W}{e}, \tag{1}$$

where h is Planck's constant, e the elementary charge, ν the frequency of the incident light, and W the work function of the metal surface. Equation (**??**) implies a *linear relationship* between V_0 and ν with slope h/e and (negative) intercept $-W/e$.

2 Pre-Einstein Experimental History

2.1 Hertz (1887) and Hallwachs (1888)

Heinrich Hertz observed in 1887 that ultraviolet light falling on a metal spark gap reduced the voltage needed to produce a spark. Wilhelm Hallwachs (1888) showed that a zinc plate illuminated by ultraviolet light discharged negative charge and acquired positive charge—i.e., light caused emission of negative particles.

2.2 Lenard's Experiments (1899–1902)

Philipp Lenard established several qualitative properties of photoelectron emission:

1. (1899) The emitted particles are identical to cathode-ray electrons, confirming J. J. Thomson's identification of the electron.

2. (1902) **Key result**: the *maximum velocity* (equivalently, the stopping potential V_0) of emitted electrons is *independent of light intensity* but depends on the frequency of the incident light. This directly contradicted classical wave theory.

3. Lenard noted that the "trigger" action of light seemed to release energy stored in the metal rather than delivering energy directly from the wave.

Quantitative limitations. Lenard's measurements of V_0 vs. ν were qualitative rather than quantitative. His metal surfaces were typically oxidized, introducing variable and uncontrolled contact potentials that shifted the apparent stopping voltages. He did not work in ultra-high vacuum. Consequently, while his *qualitative* finding (that V_0 depends on frequency, not intensity) was correct and epochal, his data could not reliably test the linear prediction (**??**). The quantitative confirmation awaited Millikan's vacuum technique developed 1908–1916.

Reference. Lenard's 1902 paper: "Ueber die lichtelektrische Wirkung," *Ann. Phys.* **313**(5):149–198 (1902). 10.1002/andp.19023130510 (Series numbering: *Ann. Phys.* **8**:149–198 in the "neue Folge" [new series].)

2.3 Other Pre-1905 Experimenters

Elster and Geitel (1889–1891). Johann Elster and Hans Geitel demonstrated that the alkali metals (Na, K, Rb, Cs) are photoelectrically active at visible wavelengths, whereas most metals require ultraviolet light. They concluded (1891) that more electropositive metals have a smaller threshold frequency ν_0 (longer threshold wavelength λ_0), anticipating point 4 of Einstein's equation. See: Wied. Ann. **43**:225 (1891).

Mikhail Stoletov (1888–1891). Aleksandr Stoletov made systematic measurements of photocurrent as a function of light intensity and polarity, establishing the photocurrent–intensity proportionality and verifying that photoelectrons carry negative charge. His apparatus closely resembles the modern photoelectric cell.

J. J. Thomson (1899). Thomson measured the charge-to-mass ratio e/m of photoelectrons and found it identical to that of cathode-ray electrons, confirming that photoelectrons are electrons.

3 Einstein's 1905 Quantum Hypothesis

In his March 1905 paper Einstein proposed that electromagnetic radiation of frequency ν consists of discrete quanta ("light quanta," later "photons") each carrying energy $h\nu$. A photoelectron absorbs one quantum and escapes the metal with kinetic energy

$$\tfrac{1}{2}m_e v_{\max}^2 = eV_0 = h\nu - W,$$

which is exactly equation (**??**). At the time, "there were available no experiments whatever for determining anything about how P.D. varies with ν" (Millikan, 1916, p. 356). Einstein's prediction was considered "bold, not to say the reckless, hypothesis" (Millikan's phrase, echoing the general consensus).

Reference. Einstein (1905): "Über einen die Erzeugung und Verwandlung des Lichtes betreffenden heuristischen Gesichtspunkt," *Ann. Phys.* **17**:132–148 (1905). 10.1002/andp.19053220607

4 Pre-Millikan Tests (1907–1914)

Several experiments tested aspects of Einstein's equation but fell short of definitive confirmation, primarily because of two systematic errors: (a) stray scattered light of shorter wavelength contaminating the stopping-potential determination, and (b) unknown contact potentials between the photoemitting surface and the collector (which shift the intercept $-W/e$ by a constant but do not affect the *slope* h/e unless they vary between measurements).

Ladenburg (1907). First quantitative measurements of V_0 vs. ν, but over a wavelength range of only 50 nm (2000–2600 Å), making it impossible to discriminate $V_0 \propto \nu$ (Einstein) from $V_0 \propto \nu^{1/2}$ (classical).

Hughes (1912). A. Ll. Hughes measured stopping potentials at three frequencies (corresponding to $\lambda = 1849$, 2257, and 2537 Å) for several metals using a refined vacuum apparatus. Two of the three points were used to fix the regression line; the third lay close to it. Using a two-frequency fit, Hughes found slope values ranging from 3.55×10^{-27} to 5.85×10^{-27} erg·s

across eight metals (compared to Planck's $h = 6.55 \times 10^{-27}$ erg·s), a discrepancy of 10–45%. Systematic errors (contact potentials unmeasured in vacuo; only three frequencies covering a 60 nm range; scattered short-wave-length light) prevented a precise determination.

Reference. Hughes (1913): "On the emission velocities of photo-electrons," *Phil. Trans. R. Soc. A* **212**:205–226 (1913). 10.1098/rsta.1913.0007

Richardson and Compton (1912). Owen Richardson and Karl Compton made measurements on eight metals. Their slopes (determining h) varied by more than 60% across metals, which Millikan attributed chiefly to the restricted wavelength range (all their curves save Na and Al were between 2100 and 2750 Å, just 65 nm wide) and to lack of contact-EMF correction for sodium. Their sodium result gave $h \approx 5.2 \times 10^{-27}$ erg·s (20% low) and their aluminum result $h \approx 4.3 \times 10^{-27}$ erg·s (35% low).

Reference. Richardson and Compton (1912): "The photoelectric effect," *Phil. Mag.* **24**:575–594 (1912). 10.1080/14786441208637039

Millikan (1914, preliminary). Millikan reported preliminary results to the American Physical Society in April 1914 (sodium data only). **Reference.** Millikan (1914): "A Direct Determination of 'h'," *Phys. Rev.* **4**(1):73–75 (1914). 10.1103/PhysRev.4.73.2

5 Millikan's Definitive Measurement (1916)

5.1 Apparatus: the "Machine Shop in Vacuo"

Millikan's key innovation was eliminating surface contamination by building what he called a "machine shop in vacuo." Inside an evacuated bulb, an electromagnetically controlled rotating knife (K) was used to cut a fresh shaving from a cast cylinder of sodium, potassium, or lithium mounted on a wheel (W). The wheel was then rotated to bring the freshly scraped surface into alignment with the monochromatic light beam. All metal surfaces, knife, and wheel were controlled entirely by external electromagnets—no mechanical feedthroughs that might cause vacuum leaks. This was the ninth tube of the general design; the mechanic Julius Pearson is explicitly credited.

The alkali metals studied were **sodium** (Na) and **lithium** (Li). (Potassium data were lost to an accident before publication.)

The light source was a Heracus 200-volt quartz-mercury arc. A Hilger monochromator selected individual spectral lines. Only spectral lines that had no companion lines on the short-wave-length side (i.e., no contamination by higher-frequency light) were used.

Stopping potentials were measured by the Kelvin method (Kelvin double-balance): the potentiometer voltage between the alkali metal and the collecting plate S was adjusted until lifting the plate produced no deflection in a connected electrometer. The electrometer sensitivity was 2500 mm/V; the Weston Standard Laboratory voltmeter used was certified by the National Bureau of Standards to 0.1%.

5.2 Mercury Arc Lines Selected

The following mercury spectral lines were used (wavelengths determined by Walter T. Whitney using a grating photograph):

Table 1: Mercury arc spectral lines used by Millikan (1916). Frequencies computed from $\nu = c/\lambda$ with $c = 2.998 \times 10^{10}$ cm/s.

λ (Å)	λ (nm)	ν (10^{14} Hz)
5461.0	546.10	5.490
4339.4	433.94	6.908
4046.8	404.68	7.409
3650.2	365.02	8.213
3125.5	312.55	9.591
2534.7	253.47	11.827
2399	239.9	12.497

5.3 Systematic Error Controls

Stray short-wave-length light. For each spectral line, filters were interposed to absorb all light of shorter wavelength. Without this precaution, faint scattered UV light from the monochromator prism and lenses would produce photoelectrons at all wavelengths, pushing the apparent stopping potential for longer (lower-energy) lines upward and flattening the slope *below* h/e. Millikan showed this was a primary reason previous workers (Ladenburg, Richardson & Compton, Hughes) obtained values of h 10–45% below Planck's value.

Reverse electronic current. If the wavelength used *exceeds* the threshold wavelength of the collecting cylinder, electrons liberated from the cylinder walls by reflected light create a "back leak" of current in the wrong direction, pushing the apparent intercept toward shorter wavelengths. Millikan used copper-oxide walls whose long-wavelength limit was $\lambda = 2688$ Å (for the sodium tube) and $\lambda \approx 2535$ Å (for the lithium tube), and corrected the line 2535 reading accordingly. Lines 3125 through 5461, which lie above the oxide threshold, were free of this error.

Contact electromotive force. The contact EMF between alkali metal and collector varied from 0.4 to 3.0 V depending on surface age and absorbed gas, and changed rapidly in the minutes after shaving. The contact EMF acts as an additional retardation or acceleration for photoelectrons. With all lines held constant in every other respect, the contact EMF shifts the entire V_0–ν curve vertically but does *not* change its slope. Thus the slope h/e can be determined accurately even if the contact EMF is not independently measured—provided the EMF is *constant* during the full series of measurements.

Millikan found that with ~ 0.01 mm Hg of residual gas in the bulb, the contact EMF stabilised within about one hour of shaving and remained constant to within ± 0.01 V for several hours—ample time for a complete set of measurements. In the main sodium run, the contact EMF was measured independently by the Kelvin method as $V_c = 2.51$ V (Na positive with respect to Cu oxide), remaining stable from 2.495 V (6 h after shaving) to 2.493 V (9 h after shaving).

5.4 Raw Data: Sodium (Table I)

Millikan tabulated the electrometer deflections (in mm, for a 30-second illumination) as a function of applied potentiometer voltage for each spectral line (Table I of the original paper). The balance point (zero deflection) gives the intercept. Deflections at the last few settings before the balance for the main sodium curves are reproduced below.

Table 2: Selected readings from Millikan (1916), Table I. Sodium surface. "Volts" is the applied potentiometer voltage; "Def'n" is the electrometer deflection in mm (30-s illumination). For lines 5461–3126 the contact EMF ($V_c = 2.51$ V) exceeds the photopotential so the balance falls on the negative-voltage side; the magnitudes of the applied *assisting* potential near balance are listed. For line 2535 the balance falls on the positive (retarding) side.

5461 Å		4339 Å		4047 Å		3650 Å		3126 Å		2535 Å	
V (V)	mm	V (V)	mm	V (V)	mm	V (V)	mm	V (V)	mm	V (V)	mm
2.257	28	1.581	44	1.576	82	1.157	67.5	0.581	52	−0.058	68
2.205	14	1.629	20	1.524	55	1.105	36	0.529	29	+0.058	38
2.152	7	1.576	10	1.471	36	1.053	19	0.477	12	+0.162	26
2.100	3	1.524	4	1.419	24	1.000	11	0.424	5	+0.267	16.5
		1.367	3	0.948	4			0.372	2.5	+0.372	8

5.5 Determination of h from Sodium

Main five-point fit (Fig. 6 of the paper). The balance-point intercepts from the curves of Fig. 5 were plotted against frequency ν. The slope of the resulting straight line, determined

primarily from the five most reliable points ($\lambda = 5461, 4339, 4047, 3650, 3126\,\text{Å}$), was

$$\left.\frac{V_0}{\nu}\right|_{\text{slope}} = 4.124 \times 10^{-15}\ \text{V·s} \quad \text{(Fig. 6 fit, sodium, Millikan 1916, p. 374).}$$

"No potential departs from the line by more than 0.01 V and since the range of volts is about 2, it is a conservative estimate to place the maximum uncertainty in the slope at about 1 in 200 or 0.5 per cent" (Millikan 1916, p. 374).

Nine two-point determinations (Table II of the paper). To check for consistency and eliminate the influence of a slowly drifting contact EMF, Millikan also measured the slope from pairs of lines on different days. Line 5461 was always one of the two (chosen for its large intensity and hence most precise intercept). The nine results are reproduced in Table **??**.

Table 3: Millikan (1916): Table II. Nine independent two-point determinations of the slope h/e from sodium. Each row is from a different day and a different vacuum condition. The "estimated uncertainties" quoted by Millikan at the time of measurement range from 1% to 4%, with lines 3650 and 3126 most reliable.

λ compared with 5461 Å	Slope ($\times 10^{-15}$ V·s)
3126	4.11
3650	4.14
3126	4.10
3650	4.12
3126	4.24
4047	3.98
2535	4.04
3126	4.24
4047	4.21
Mean	**4.131**

Combined sodium result. Taking the mean of 4.124 (from the five-point fit) and 4.131 (from Table **??**), Millikan obtained the combined sodium slope:

$$\frac{h}{e} = 4.128 \times 10^{-15}\ \text{V·s}.$$

With his oil-drop value $e = 4.774 \times 10^{-10}$ esu $= 1.592 \times 10^{-19}$ C this gives

$$h_{\text{Na}} = 4.128 \times 10^{-15} \times 1.592 \times 10^{-19} = 6.572 \times 10^{-34}\ \text{J·s} \approx 6.57 \times 10^{-27}\ \text{erg·s}.$$

5.6 Lithium Results

For lithium the measurements were easier: the photosensitivity remained high for months without reshaving. Two independent series gave:

First lithium series (Table III of the paper). Contact EMF = 1.52 V; threshold frequency $\nu_0(\text{Li}) = 57.0 \times 10^{13}$ Hz ($\lambda_0 \approx 5260$ Å). At $\lambda = 2535$ Å: $V_{0,\text{obs}} = 1.00$ V, so $V_{0,\text{true}} = 1.00 + 1.52 = 2.52$ V.

Second lithium series (six months later, Fig. 9 of the paper). Contact EMF = 1.11 V; $\nu_0(\text{Li}) = 59.7 \times 10^{13}$ Hz ($\lambda_0 \approx 5020$ Å). At $\lambda = 2535$ Å: $V_{0,\text{obs}} = 1.29$ V, so $V_{0,\text{true}} = 1.29 + 1.11 = 2.40$ V.

From the two lithium series, Millikan obtained:

$$h_{\text{Li}} = 6.584 \times 10^{-27} \text{ erg·s} \quad \text{(mean of two determinations; uncertainty} \approx 1\%\text{)}.$$

5.7 Final Result

Placing greater reliance on the sodium determination (smaller uncertainty):

$$\boxed{h = 6.57 \times 10^{-27} \text{ erg·s} = 6.57 \times 10^{-34} \text{ J·s}} \quad \text{(precision } \approx 0.5\%\text{)}. \tag{2}$$

The modern (NIST CODATA 2022) value is $h = 6.626\,070\,15 \times 10^{-34}$ J·Hz^{-1} (exact by SI definition since 2019). Millikan's result is low by about 0.8%—well within his stated uncertainty—attributable primarily to the small error in his oil-drop value of e (Millikan used $e = 1.592 \times 10^{-19}$ C; the modern value is $e = 1.602\,176\,634 \times 10^{-19}$ C, about 0.65% higher).

6 Reconstructed (ν, V_0) Data Table for Linear Regression

The *true* stopping potentials in Einstein's equation (**??**) are $V_{0,\text{true}} = V_{0,\text{obs}} + V_c$, where V_c is the contact EMF, which shifts the curve vertically but does not affect the slope. These true stopping potentials can be reconstructed from the known slope $h/e = 4.128 \times 10^{-15}$ V·s, the independently determined threshold frequency ν_0, and the explicitly stated $V_{0,\text{true}}$ for the line 2535 Å.

The relevant parameters are:

$$\begin{aligned}
\nu_0(\text{Na}) &= 4.39 \times 10^{14}\ \text{Hz} \quad (\lambda_0 = 6800\,\text{Å, from contact-EMF analysis}),\\
\nu_0(\text{Li}) &= 5.70 \times 10^{14}\ \text{Hz} \quad (\lambda_0 = 5260\,\text{Å, first series}),\\
W_{\text{Na}} &= h\nu_0 = 6.57 \times 10^{-27} \times 4.39 \times 10^{13} = 2.88 \times 10^{-12}\ \text{erg} \approx 1.80\ \text{eV},\\
W_{\text{Li}} &= 6.57 \times 10^{-27} \times 5.70 \times 10^{13} = 3.74 \times 10^{-12}\ \text{erg} \approx 2.34\ \text{eV}.
\end{aligned}$$

Table 4: Reconstructed true stopping potentials V_0 for sodium and lithium, computed from $V_0 = (h/e)(\nu - \nu_0)$ with $h/e = 4.128 \times 10^{-15}$ V·s. For sodium: $\nu_0 = 4.39 \times 10^{14}$ Hz. For lithium: $\nu_0 = 5.70 \times 10^{14}$ Hz (first series). The explicit data point for Na at 2535 Å is stated in the paper as $V_{0,\text{obs}} = 0.52$ V and $V_c = 2.51$ V, giving $V_{0,\text{true}} = 3.03$ V (computed: 3.07 V; the small difference reflects scattered-light correction for this line noted by Millikan).

λ (Å)	λ (nm)	ν (10^{14} Hz)	V_0(Na) (V)	V_0(Li) (V)
5461.0	546.10	5.490	0.454	—[a]
4339.4	433.94	6.908	1.039	0.499
4046.8	404.68	7.409	1.246	0.706
3650.2	365.02	8.213	1.578	1.037
3125.5	312.55	9.591	2.147	1.606
2534.7	253.47	11.827	3.07[b]	2.529

Line 5461 lies above the long-wavelength limit of the lithium surface (Li threshold $\lambda_0 \approx 5260$ Å), so lithium is not photoelectrically active at 5461 Å.

The paper states the *observed* $V_{0,\text{obs}} = 0.52$ V with $V_c = 2.51$ V, giving $V_{0,\text{true}} = 3.03$ V. The computed value 3.07 V from $h/e \cdot (\nu - \nu_0)$ is consistent within the stated 1% uncertainty for this line.

Linear regression. A linear regression of V_0 on ν using the five most reliable sodium data points gives:

- Slope: $h/e = 4.128 \times 10^{-15}$ V·s
- Intercept: $-W_{\text{Na}}/e = -1.80$ V (at $\nu = 0$)
- Implied h (with modern e): $4.128 \times 10^{-15} \times 1.602 \times 10^{-19} = 6.61 \times 10^{-34}$ J·s (0.5% high relative to modern value due to rounding)

Long-wavelength limit and work function. Extrapolation of the fitted line to $V_0 = 0$ gives ν_0 (threshold frequency). From the contact-EMF-corrected line for sodium, Millikan

found $\nu_0 = 43.9 \times 10^{13}$ Hz, corresponding to $\lambda_0 = 6800$ Å (deep red). Direct measurement confirmed that sodium was photoelectrically active at the yellow mercury lines $\lambda = 5780, 5790$ Å and even at the red line $\lambda = 6235$ Å, consistent with $\lambda_0 \approx 6800$ Å.

7 Millikan's Reluctance to Accept the Photon

A remarkable feature of this episode is that Millikan, while confirming Einstein's equation to within 0.5%, refused to accept the physical interpretation. He wrote in January 1916 (before the definitive result):

> "Einstein's photoelectric equation ... cannot in my judgment be looked upon at present as resting upon any sort of a satisfactory theoretical foundation."

Yet the equation "actually represents very accurately the behavior." In his 1916 paper he describes Einstein's light-quantum hypothesis as "bold, not to say the reckless, hypothesis."

By 1923 (after Compton scattering), Millikan had accepted the photon. In his 1923 Nobel address he said: "This work resulted, contrary to my own expectation, in the first direct experimental proof ... of the Einstein equation" (quoted by Holton, 1999).

8 Circuit of Prior Results and Millikan's Innovations

Table 5: Comparison of key pre-Millikan photoelectric measurements of h/e (or equivalently of h using the known e). Millikan attributes the systematic underestimation in earlier work to stray short-wavelength light, restricted frequency range, and uncontrolled contact EMFs.

Authors	Year	Metal(s)	h obtained (10^{-27} erg·s)	Error
Hughes	1912	Cu, Zn, Al, Sn, Pb, etc.	3.55–5.85	−10 to −45%
Richardson & Compton	1912	8 metals incl. Na, Al	∼3.5–5.8	large
Millikan (Na)	1916	Sodium	6.569	−0.8%
Millikan (Li)	1916	Lithium	6.584	−0.6%
Planck (radiation)	1913	—	6.415	−3.2%
Millikan mean	1916	Na + Li	**6.57**	**−0.84%**
Modern (NIST 2022)	2022	—	6.6261	0 (exact)

9 Key References

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10 Physical Constants Used

Table 6: Physical constants. Millikan's values (1916) vs. modern NIST CODATA 2022. The 0.65% error in Millikan's e propagates directly to his h.

Constant	Millikan (1916)	Modern (NIST 2022)	Error
h (erg·s)	6.57×10^{-27}	6.6261×10^{-27}	-0.84%
e (esu)	4.774×10^{-10}	4.803×10^{-10}	-0.60%
e (C)	1.592×10^{-19}	$1.602\,176\,634 \times 10^{-19}$ (exact)	-0.65%
h/e (V·s)	4.128×10^{-15}	4.136×10^{-15}	-0.19%